MILITARY SPECIFICATION

CRADLES, WEAPON, SHIPPING AND STORAGE, MK 12, MK 13, MK 14, MK 15, MK 16 AND MK 18

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers Mk 12, Mk 13, Mk 14, Mk 15, Mk 16 and Mk 18 Weapon Cradles, for shipping and storage of completely assembled weapons (less the removable aerodynamic control surfaces). The cradles are designed in multiple-round configurations (see 6.1).

1.2 Classification - Cradles covered by this specification shall be of the following designations, as specified (see 6 2)

Mk 12 Cradle, Guided Missile (SPARROW III) (AIM-7D and 7E)
Mk 13 Cradle, Guided Weapon (WALLEYE)
Mk 14 Cradle, Guided Missile (SHRIKE AGM-45A)
Mk 15 Cradle, Guided Missile (BULLPUP AGM-12B)
Mk 16 Cradle, Guided Missile (SIDEWINDER AIM-9C and -9D)
Mk 18 Cradle, Weapon (ROCKEYE II CLUSTER BOMB)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

FSC 8140

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.

Military (Continued)

MIL-STD-130	Identification Marking of U.S Military Property

DRAWINGS

Naval Air Systems Command

DL 67A97H1	Cradle, Guided Missile, Mk 12 Mod 0, Drawing List
DL 67A101H1	Cradle, Guided Weapon, Mk 13 Mod 0, Drawing List
DL 67A102H1	Cradle Guided Missile, Mk 14 Mod 0, Drawing List
DL 2580883	Cradle, Guided Missile, Mk 15 Mod 0, Drawing List
DL 2580887	Cradle, Guided Missile, Mk 16 Mod 0, Drawing List
DL 2581189	Cradle, Weapon, Mk 18 Mod 0, Drawing List

PUBLICATIONS

Naval Ordnance Systems Command Weapons Requirement

WR 54/168	Weapons Requirement, Palletizing Fleet Issue Unit Load, Bomb Cluster Mk 20 Mod 2 (Rockeye II) in Cradle, Weapon Mk 18 Mod 0
WR 54/169	Weapons Requirement, Palletizing Fleet Issue Unit Load, Sidewinder Missile AIM-9C and AIM-9D (less wings and fins) in Cradle, Weapon Mk 16 Mod 0
WR 54/171	Weapons Requirement, Palletizing Fleet Issue Unit Load, Sparrow III Missile, AIM-7D and AIM-7E (less wings and fins) in Cradle, Weapon Mk 12 Mod 0, DL 67A97H1
WR 54/172	Weapons Requirement, Palletizing Fleet Issue Unit Load, SHRIKE Missile, AGM-45A (less wings and fins) in Cradle Weapon Mk 14 Mod 0, DL 67A102H1
WR 54/182	Weapons Requirement, Palletizing Fleet Issue Unit Load, Bullpup Missile AGM-12B (less wings and fins) in Cradle, Weapon Mk 15 Mod 0, DL 2580883

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government quality assurance representative Copies of this specification and other unclassified specifications and standards required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Publications and Forms Center (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All documents, including drawings, not available from the above source should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article - The cradles furnished under this specification shall be products which have been inspected and passed the first article inspection specified herein

3.2 General - Cradles covered by this specification shall be manufactured in accordance with the Naval Air Systems Command drawings and the performance requirements specified herein.

3.3 Conflicting requirements - Conflicting requirements arising between this specification or any specification, publication, or drawing listed herein, shall be referred in writing to the procuring agency or contracting office for interpretation and classification.

3.4 Construction -

3.4.1 Mk 12 cradle - Mk 12 cradles shall be constructed in accordance with the drawings detailed in Naval Air Systems Command Drawing List DL 67A97H1 and associated parts list, and shall accommodate three SPARROW AIM-7D or AIM-7E guided missiles. (See 6.1)

3.4.2 Mk 13 cradle - Mk 13 cradles shall be constructed in accordance with the drawings detailed in Naval Air Systems Command Drawing List DL 67A101H1 and associated parts list, and shall accommodate two WALLEYE weapons. (See 6 1.)

3.4.3 Mk 14 cradle - Mk 14 cradles shall be constructed in accordance with the drawings detailed in Naval Air Systems Command Drawing List DL 67A102H1 and associated parts list, and shall accommodate three SHRIKE AGM-45A guided missiles. (See 6.1.)

3.4.4 Mk 15 cradle - Mk 15 cradles shall be contructed in accordance with the drawings detailed in Naval Air Systems Command Drawing List DL 2580883 and associated parts list, and shall accommodate two BULLPUP AGM-12B guided missiles. (See 6.1.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.

3.4.5 Mk 16 cradle - Mk 16 cradles shall be constructed in accordance with the drawings detailed in Naval Air Systems Command Drawing List DL 2580887 and associated parts list, and shall accommodate four SIDEWINDER AIM-9C or AIM-9D guided missiles. (See 6.1.)

3.4.6 Mk 18 cradle - Mk 18 cradles shall be constructed in accordance with the drawings detailed in Naval Air Systems Command Drawing List DL 2581189 and associated parts list, and shall accommodate two ROCKEYE II Cluster Bombs. (See 6.1.)

3 4.7 Construction details - The components noted below, where called for in the detail drawings, shall meet the following requirements.

3.4 7.1 Gaskets - Gaskets for closure of cover assemblies shall be fabricated from extruded stock as indicated on the applicable drawings. The joined ends shall be vulcanized to form a smooth, strong pliable joint. The length of the fabricated gaskets shall be such as to maintain the pressure specified in 3.5.1.

3.4.7.2 Gasket grooves - The entire length, including curved sections of gasket grooves, shall be uniform in cross-section, free of cracks, seams, porosity, inclusions, or any other defects which, due to their nature may be the cause of air leakage between the shells and gasket

3 4 7 3 Sealing rails - The female sealing rail in Mk 12, 13, and 14 shall be formed by bending and butt welding of the ends Inserts or some other suitable method shall be used when bending, to prevent buckling and distortion of the cross sectional shape of the gasket cavity. The ends of the extrusions shall be joined to obtain a smooth, continuous surface, free of weld splatter, excessive weld, voids and excessive marks due to smoothing of the welds. The length and width of sealing rails shall be so fabricated that when cover and base are assembled, the container shall meet the pressure retention requirement specified in 3 5.1.

3 4.7 4 Clamp assemblies - Clamp assemblies in Mk 15, 16 and 18 after installation on the cradles shall be easily installed and removed from the shell assembly. With the corresponding latch installed on the clamp assembly, and in a closed position attached to the pod assembly, the gasket shall then be depressed a minimum of 0.125 inch. Free movement between the clamp, upper and lower, after assembly with the shouldered rivet shall be possible.

3.4.7 5 Saddle assembly - Each detail part comprising the saddle in Mk 15, 16 and 18 cradles shall be in T-6 temper condition prior to weldment. Care shall be exercised in assembly as to requirements of flatness, parallelism and squareness.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.

3.4.7.6 Shell straps - Each strap in Mk 15, 16 and 18 cradles, after installation on the shell assembly shall be in perfect alignment with the latch assembly installed on the saddle assembly.

3.4.7.7 Elastomeric shock mounts - Shock mount suspension shall be used to carry the load of the weapon and container case or pads where required. The mounts shall have a transmissibility factor of five (5) or less. Shock mounts shall operate successfully in the temperature range of -65° F to +160° F.

3.4.7.7.1 Elastomeric shock mounts shall be certified by the source or supplier as complying with the performance requirements of this specification, and shall conform with the detailed drawings for each cradle, where applicable.

3.5 Performance requirements and product characteristics -

3.5.1 Pressure retention - The rate of pressure decrease shall not exceed 1.5 inches of water (0.05 psig/hr) when the completely assembled cradle is pressurized to a minimum positive pressure of 13.85 +0.15, -0.00 inches of water (0.50 psig), for one hour.

3.5.2 Drop test (free fall) - Each cradle loaded with dummy weapons shall be subjected to the free fall drop test as described in Section 4 and shall then pass the pressure retention test (3.5.1).

3.5.3 Vibration - When subjected to the test in Section 4, each container shall provide suitable protection for its contents. The contents shall not be damaged or shifted so as to reduce the utility of the contents or the container. In addition, no individual suspension system used in the cradle shall exhibit a natural frequency in the 2 to 5 cps frequency range and the transmissibility factor shall not exceed 5 with an allowable increase during 45 minutes to 6.5. However, recovery shall be 90 percent after 2 hours allowable return to ambient.

3.5.4 Hoisting - Each cradle loaded with dummy weapons of the proper weight shall be suspended by one of its lifting eyes or bails, as applicable. Each eye or bail shall be individually tested as described in 4.6.4. The bails, clamps or eyes shall not fracture or show macroscopic evidence of structural weakness.

3.5.5 Temperature test - Each cradle shall withstand the temperature conditions specified in 4.6.5 and thereafter shall comply with the pressure retention requirement specified in 3.5.1.

3.5.6 Stacking - Each complete cradle assembly of any one design shall be capable of stacking one upon the other in all combinations. The stacking posts shall slide over the lower cradle posts without interference, and while in the stacked position shall permit engagement of the referenced lock pin applicable to the particular cradle design through each of the stacking posts. (See 4.6.6.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.

3.6 Workmanship - It shall be the manufacturer's responsibility to perform the required welding, bonding and assembly procedures in such a manner that the enclosure (base assembly and cover), or shell assembly, will maintain a leak rate no greater than that specified in 3.5.1. The workmanship exhibited in the first article samples will be evaluated to determine acceptability The approved standards of workmanship will thereby become a minimum production requirement for all units offered for acceptance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The examination and testing of the cradles shall be classified as follows:

(a) First article inspection -

(1) Source inspection - Source inspection shall consist of purchase item examinations and tests required to certify the article for use, subject to satisfactory performance under first article inspection.

(2) Initial production inspection - Initial production inspection consists of examinations and tests performed on cradles which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification (see 4.3).

(b) Quality conformance inspection - Quality conformance inspection consists of examinations and tests performed on individual cradles in production lots to determine conformance of the cradles with requirements set forth in this specification (see 4.4).

4.3 First article inspection -

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.

4.3.1 First article samples - Unless otherwise specified in the contract or order, first article inspection samples shall consist of two cradles of the same design, configuration and Parts or Data List number, manufactured using the same methods, procedures, and processes, and at the same location as proposed for full production.

4.3.2 First article inspection and tests - The contractor shall perform such quality conformance tests as would be necessary to ensure conformance of production cradles to the requirements of this specification. For the purpose of first article acceptance the two sample cradles shall be subjected to the tests in Table I and in the sequence shown. Dimensional and material inspection are further detailed in 4.4.2. Upon conditional acceptance by the inspector, the first article samples shall be forwarded to an activity designated in the contract or by the procuring activity (6.2), for the purpose of determining compliance with all requirements of the contract, this specification, and the drawings Further production of the cradle by the contractor prior to approval of the contracting activity shall be at the contractor's risk. Acceptance shall be made on an individual basis (cradles of the same design) with no defects in any of the cradles submitted.

TABLE I

SAMPLE SIZE, AND SEQUENCE OF TESTS
FOR FIRST ARTICLE INSPECTION

Sample size	Sequence for sample cradle No. 1	Sequence for sample cradle No. 2
2 cradles	4.4.2 Dimensional and material inspection in accordance with applicable construction details as shown on drawing	4.4.2 Dimensional and material inspection in accordance with applicable construction details as shown on drawing
	4.6.1 Pressure retention	4.6.1 Pressure retention
	4.6.2 Free fall drop	4.6.5 Temperature test
	4.6.3 Vibration	
	4.6.1 Pressure retention	4.6.1 Pressure retention

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.

TABLE I (Continued)

Sample size	Sequence for sample cradle No. 1	Sequence for sample cradle No. 2
2 cradles (Cont'd)	4.6.4 Hoisting test	4.6.6 Stacking test
	4.6.6 Stacking test	
	ACCEPTANCE OR REJECTION	ACCEPTANCE OR REJECTION

4.4 Quality conformance inspection and test of production cradles - The quality conformance inspection shall consist of the following inspection and test.

	Paragraph
Dimensional and material inspection	4.4.2
Pressure retention test	4.6.1

4.4.1 Lot and Acceptable Quality Level - All cradles of the same design and configuration, manufactured under the same conditions and presented for acceptance at one time shall be considered an inspection lot for quality conformance tests. Inspection shall be conducted in accordance with MIL-STD-105 at Level S-4 with an Acceptable Quality Level (AQL) of 1.0 defective. A cradle is defective if it fails to meet any of the dimensional, material or test requirements.

4.4.2 Dimensional and material requirements - The classification of defects shall be as follows:

CLASSIFICATION OF DEFECTS

Containers not in conformance with applicable design drawing(s) accepted by the Government under contract (for details not specifically enumerated below). (Verify presence of all components.)

EXAMINE	DEFECT
Gaskets	Not from extruded stock, as specified. (Verification to be made during construction.) Joined ends not in smooth pliable joint. Insufficient length.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS (Continued)

EXAMINE	DEFECT
Gasket grooves	Not uniform in cross-section in straight and curved sections. Cracks present. Evidence of porosity or inclusions. Presence of defects which will cause air leakage.
Sealing rails (Mk 12, 13 and 14 only)	Buckled, distorted or presence of conditions which will cause buckling of cross-sectional shape cavity. Ends of extrusions not joined in smooth continuous weld, presence of weld splatter, excessive weld, and presence of inferior workmanship due to smoothing of welds.
Clamp assemblies (Mk 15, 16 and 18 only)	Clamp assemblies do not install on the shell assembly. Gasket does not compress at least 0.125 inch.
Saddle assembly (Mk 15, 16 and 18 only)	Temper of detail parts of assembly not T-6 prior to weldment. Parallelism, flatness and squareness unsatisfactory.
Shell strap (Mk 15, 16 and 18 only)	Not in perfect alignment with saddle assembly.

4.4.3 Test equipment and test facilities - Unless otherwise specified, the manufacturer or commercial laboratory (see 4.1) shall furnish and maintain all normal test equipment, facilities, and personnel necessary for performing all quality conformance tests. The test equipment shall be adequate in quantity, and unless definite requirements are specified, they shall be of sufficient accuracy and quality to permit performance of the required quality conformance test.

4.4.3.1 Water manometer - A water manometer capable of measuring a pressure of 15 inches of water with an overall accuracy of ±0.25 inch of water and pressure differences to a sensitivity of 0.10 inch of water shall be used when performing pressure retention test of 4.6.1.

4.5 Test conditions - Unless otherwise specified in a particular test, all tests shall be performed at a temperature between 65° F and 95° F with ambient humidity.

4.6 Tests -

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.

4.6.1 Pressure retention test - Each cradle shall be tested for the requirements of 3 5 1 by the following method:

4.6.1.1 Pressurization method -

(a) Each cradle being leak tested shall be conditioned at ambient conditions for at least four hours prior to performing the leak test. The time that the cradle and all processing materials have been maintained at ambient conditions prior to or during the processing period may be considered as part of the conditioning time.

(b) Using the hole designed for the humidity indicator or the pressure relief valve, install a Schrader type tire valve using the necessary adapters. Attach a water manometer to the cradle cover or shell.

(c) Pressurize the empty cradle to 13.85 +0.15, -0.00 inches of water (0.50 psig).

(d) Record the temperature of the air surrounding the cradle being tested. Record the height of the water column when the ambient air temperature does not fluctuate more than 1° F, for a period of five minutes.

(e) At the end of one (1) hour record the pressure in the cradle by recording the differential height of the water in the manometer. Record the temperature. Subtract 0.75 inch of water for every 1° F above the original temperature and add 0.75 inch of water for every 1° F below the original temperature. A pressure decrease greater than 1.5 inches of water (from the corrected value) shall constitute an unacceptable cradle.

4.6.2 Drop test (free fall) - To determine compliance with 3.5.2, a weapon cradle, loaded with an inert dummy, shall be subjected to an 18-inch free fall flat drop test and the shock transmitted to the dummy shall not exceed 20 G. Instrumentation accelerometers shall be so mounted to record the vertical shock at the center of gravity of the weapon and at the guidance section of the weapon. This test shall be conducted once at ambient temperature, and shall be repeated immediately after the cradle has been stored in a low temperature chamber at -40 ±5° F for 2 hours. The cradle shall be examined after the test for cracks in welds and other structural damage. Minor deformation is allowed but no deformation shall have occurred that will not permit ready disassembly, reassembly and reuse of the cradles. In the event that visible damage is present to the extent that the cradle will apparently not meet the pressure retention test, additional tests shall not be conducted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.

4.6.3 Vibration test - Prior to the vibration test, the shock mounts shall be deflected one inch to determine the actual static spring rate for the mounts used. This data shall be provided in the test report. Scanning with a minimum input of 1/4 G, shall be performed to determine the natural frequency with the container secured to the testing device. Vibration dwell testing at the resonant frequency (only in the 2 to 60 Hz range) shall be of 45 minutes duration. The input frequency of vibration to the container shall be adjusted at five minute intervals to compensate for drifting due to frictional heating effects. The transmissibility shall be measured two hours after the completion of the 45 minute dwell tests to determine recoverability characteristics of the mounts. The forcing vibration movement shall be as follows:

Natural frequency in the range	Magnitude of forcing vibration (double amplitude)
up to 15 cps	0.125 inch
16 to 25 cps	0.040 inch
26 to 35 cps	0.030 inch
36 to 60 cps	0.020 inch

This test shall be repeated immediately after maintaining the container and its contents at -40° F for two hours. The cradle shall also pass the pressure retention test of 4.6.1. A leak rate greater than 1.5 inches of water/hour shall be cause for rejection.

4.6.4 Hoisting test - Using a wire rope, 3/8 inch in diameter or more, hoist the loaded cradle by one bail or lifting eye until the cradle is free of any other space supports. The cradle shall be suspended for a minimum period of two minutes. Each of the four bails or lifting eyes shall be tested in this manner

4.6.5 Temperature test - The cradle assembly closed as for shipment but not containing the weapon or inert replacement, shall be placed in a temperature test chamber large enough to provide adequate circulation of air around the cradle. The temperature of the chamber shall be regulated so that the cradle assembly is subjected to the following conditions for three complete cycles:

(a) Ambient to -40° F	time as required
(b) Hold at -40° F	2 hours
(c) Raise to +135° F	time as required
(d) Hold at +135° F	2 hours

After the three cycles, and after return to ambient temperature, the cradle shall pass the pressure retention test of 4.6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.

4.6 6 Stacking test - The cradle assemblies shall be stacked as necessary to determine conformance with 3.5.6.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing - Cradles shall be packaged (unitized) in accordance with Weapon Requirements WR 54/171, 54/172, 54/182, 54/169 and 54/168 for Mk 12, Mk 14, Mk 15, Mk 16 and Mk 18 cradles respectively. Mk 13 cradles require no special unitization.

5.2 Marking -

5.2.1 In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with MIL-STD-129.

5.2.2 Identification marking shall be in accordance with MIL-STD-130. Upper and lower shells and frames shall be marked. Additional marking shall be accomplished as required by the applicable drawing.

6. NOTES

6.1 Intended use - The cradle type containers are intended to provide an improved capability for bulk handling of weapons during the logistics cycle with resulting greater handling and stowage efficiency on aircraft carriers. Also, the cradles will provide a level of environmental protection during stowage which has not previously been afforded air launched weapons. The number of rounds per cradle are as follows (see 1.2).

Cradle Designation	Capacity of rounds per cradle	See sketch
Mk 12	3 SPARROW III Type Guided Missiles	Figure 1
Mk 13	2 WALLEYE Guided Weapons	Figure 2
Mk 14	3 SHRIKE AGM-45A Guided Missiles	Figure 3
Mk 15	2 BULLPUP AGM-12B Guided Missiles	Figure 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.

Cradle Designation	Capacity of rounds per cradle	See sketch
Mk 16	4 SIDEWINDER AIM-9C or AIM-9D Guided Missiles	Figure 5
Mk 18	2 ROCKEYE II Cluster Bombs	Figure 6

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Mark designation of cradle being purchased.

(c) Completeness and availability of dimensional, material and acceptance test data.

(d) Location of first article testing and estimated time for completion of first article testing.

6.3 Supersession data - This specification supersedes and includes the requirements of Naval Air Systems Command Purchase Descriptions WS4663, XAS-1539, XAS-1552 and XAS-1553

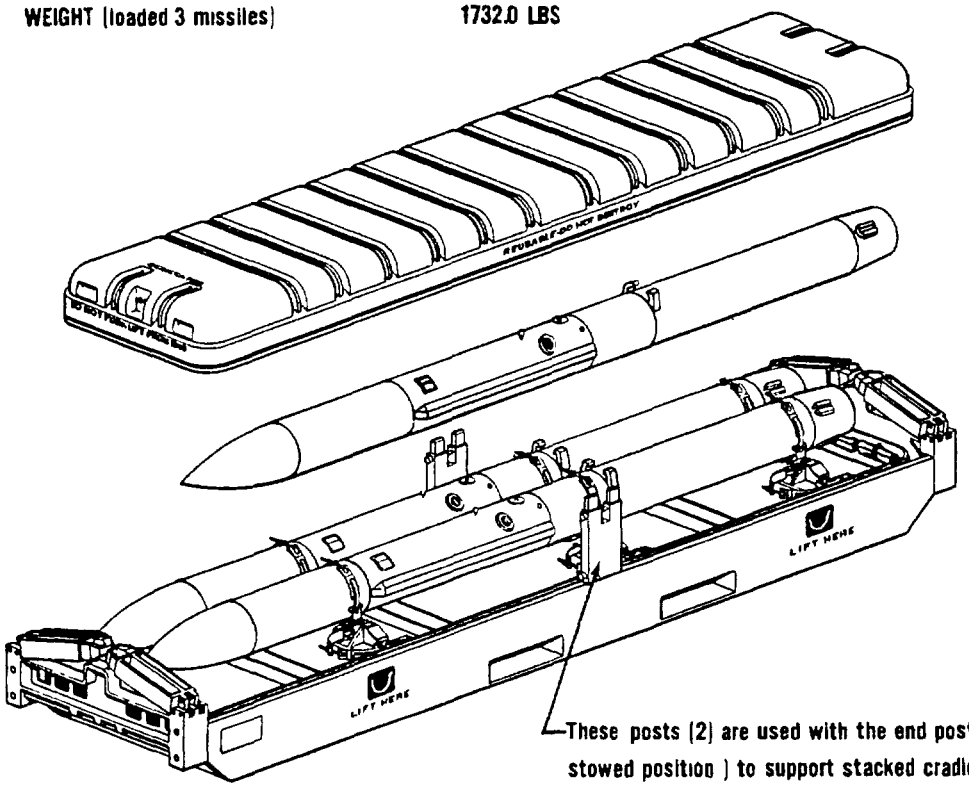
Preparing activity:
Navy - AS
(Project No. 8140-N050)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.

DATA:

GENERAL ARRANGEMENT DWG NO 67A97H1
FEDERAL STOCK NO (to be assigned)

CUBE	80 9 CU. FT
LENGTH	159 38 INCHES
WIDTH	36 0 INCHES
HEIGHT (with stacking posts up)	24.38 INCHES
HEIGHT (with stacking posts folded)	19 25 INCHES
STACKING HEIGHT (3 high)	60 75 INCHES
WEIGHT (empty)	550 0 LBS
WEIGHT (loaded 3 missiles)	1732.0 LBS



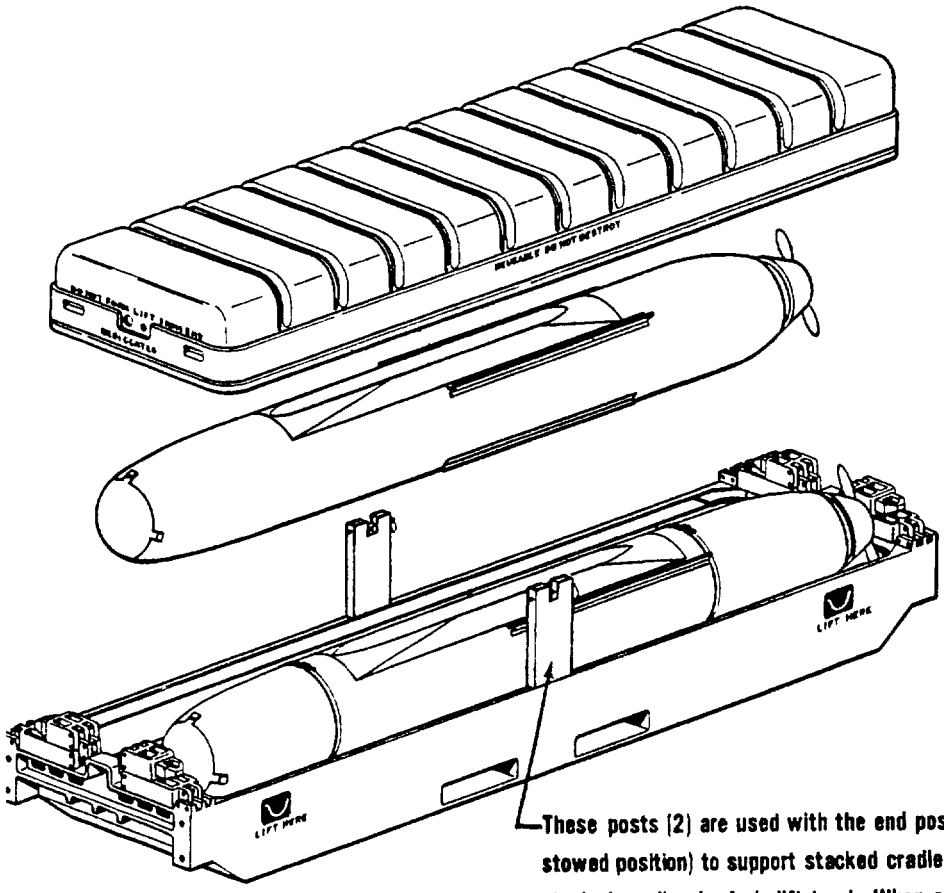
These posts (2) are used with the end posts (shown in stowed position) to support stacked cradles and when moving stacked cradles by fork lift truck When empty cradles are stored or shipped, the side posts are placed inside the cradle pod

FIGURE 1. CRADLE, GUIDED MISSILE, MK 12 MOD 0

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.

DATA:

GENERAL ARRANGEMENT DWG NO 67A101H1	
FEDERAL STOCK NO (To be assigned)	
CUBE	88.4 CU. FT.
LENGTH	150.0 INCHES
WIDTH	38.0 INCHES
HEIGHT (with stacking posts up)	33.6 INCHES
HEIGHT (with stacking posts folded)	26 8 INCHES
STACKING HEIGHT (2 high)	56.8 INCHES
WEIGHT (empty)	725.0 LBS.
WEIGHT (loaded-2 weapons)	2975 0 LBS.



These posts (2) are used with the end posts (shown in stowed position) to support stacked cradles and when moving stacked cradles by fork lift truck When empty cradles are stored or shipped, the side posts are placed inside the cradle pod

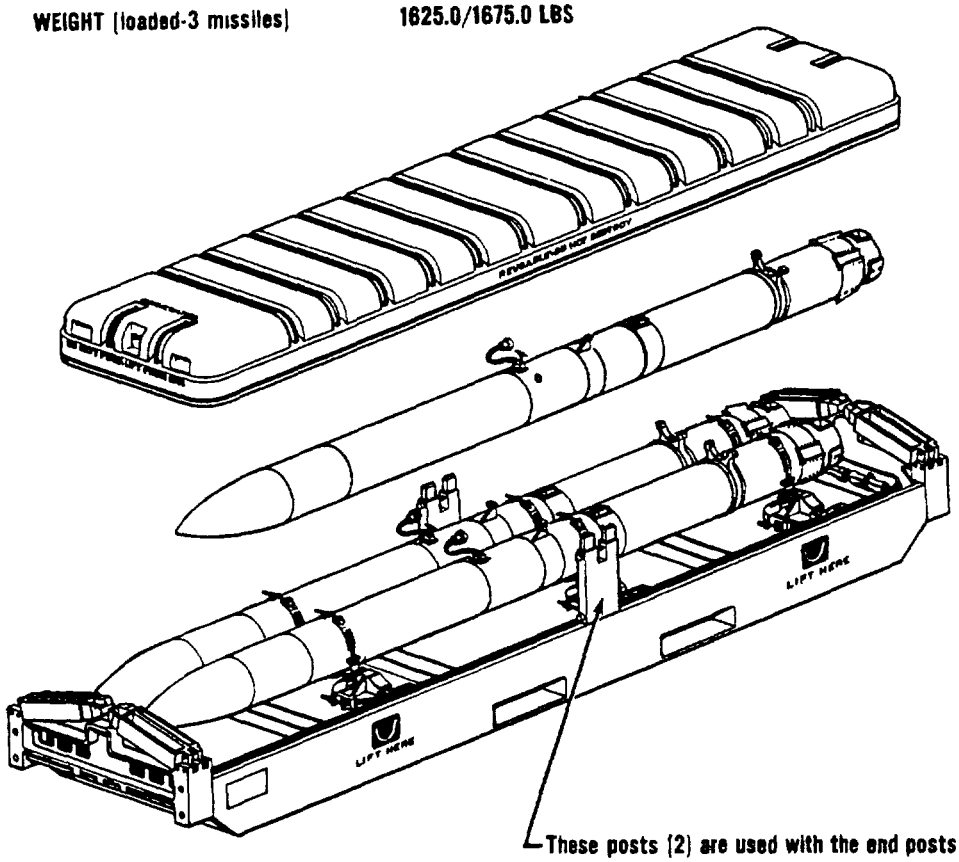
FIGURE 2. CRADLE, GUIDED WEAPON, MK 13 MOD 0

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.

DATA:

GENERAL ARRANGEMENT DWG NO 67A102H1
FEDERAL STOCK NO (to be assigned)

CUBE	56 4 CU FT.
LENGTH	140 63 INCHES
WIDTH	36.0 INCHES
HEIGHT (with stacking posts up)	24.38 INCHES
HEIGHT (with stacking posts folded)	19 25 INCHES
STACKING HEIGHT (3 high)	60.75 INCHES
WEIGHT (empty)	500.0 LBS
WEIGHT (loaded-3 missiles)	1625.0/1675.0 LBS



These posts (2) are used with the end posts (shown in stowed position) to support stacked cradles and when moving stacked cradles by fork lift truck When empty cradles are stored or shipped, the side posts are placed inside the cradle pod.

FIGURE 3. CRADLE, GUIDED MISSILE, MK 14 MOD 0

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.

DATA:

GENERAL ARRANGEMENT DWG. NO. 2580883	
FEDERAL STOCK NO. (to be assigned)	
CUBE	84.4 CU. FT.
LENGTH	153.0 INCHES
WIDTH	36.0 INCHES
HEIGHT (with stacking posts up)	26.5 INCHES
STACKING HEIGHT (3 high) TO TOP OF POST	73.0 INCHES
WEIGHT (empty)	558.0 LBS.
WEIGHT (loaded-2 missiles:	
Solid Fuel	1710.0 LBS.
Liquid Fuel)	1660.0 LBS.

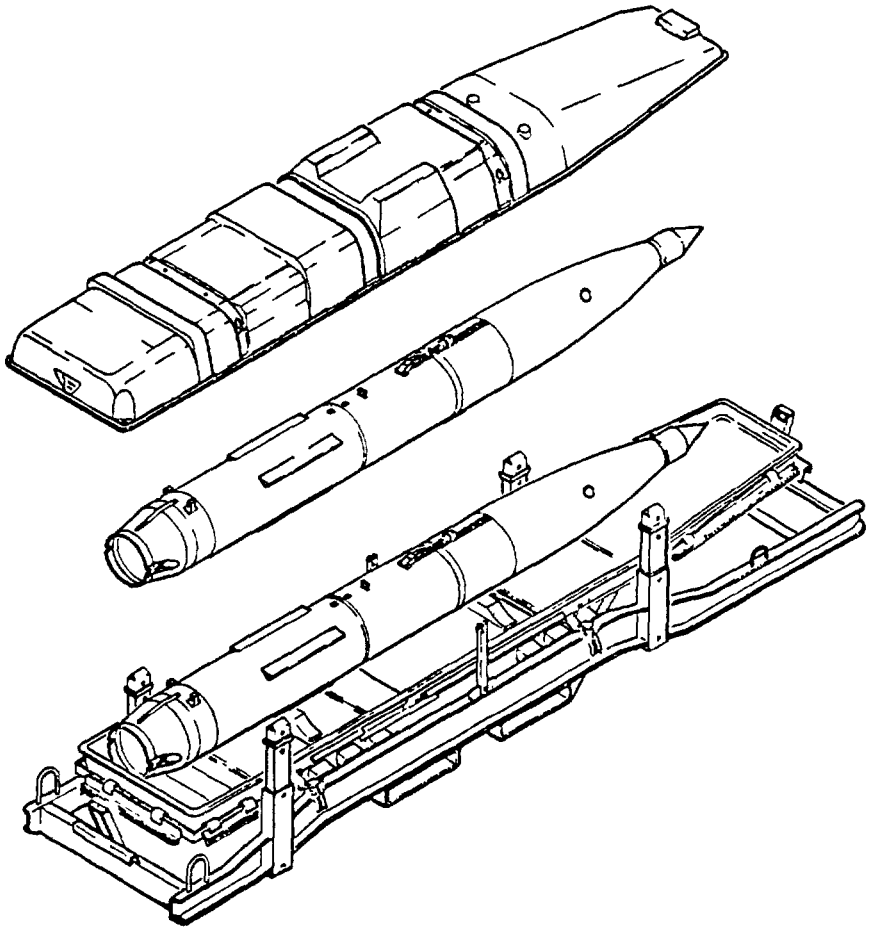


FIGURE 4. CRADLE, GUIDED MISSILE, MK 15 MOD 0

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.

DATA:

GENERAL ARRANGEMENT DWG. NO 2580887	
FEDERAL STOCK NO (To be assigned)	
CUBE	46.0 CU FT.
LENGTH	136.0 INCHES
WIDTH	35.375 INCHES
HEIGHT (with stacking posts up)	18.625 INCHES
STACKING HEIGHT (4 high) TO TOP OF POST	64.75 INCHES
WEIGHT (empty)	500.0 LBS.
WEIGHT (loaded – 4 AIM-9C)	1232.0 LBS.
WEIGHT (loaded – 4 AIM-9D)	1168.0 LBS

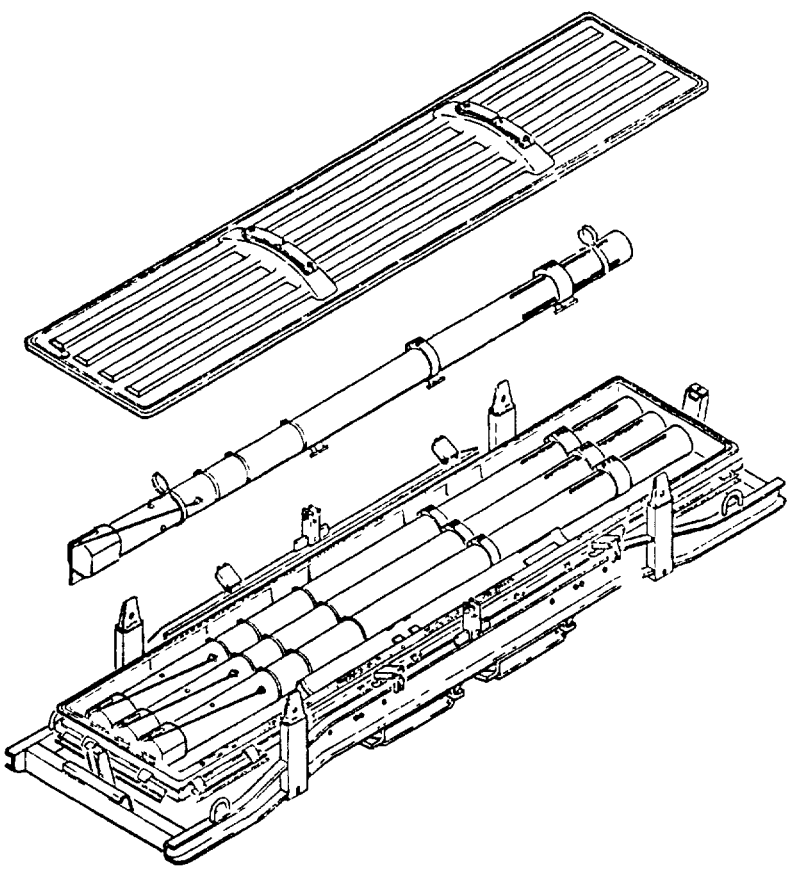


FIGURE 5. CRADLE, GUIDED MISSILE, MK 16 MOD 0

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.

DATA:

GENERAL ARRANGEMENT DWG. NO. 2581189
FEDERAL STOCK NO. (to be assigned)

CUBE	58.6 CU FT.
LENGTH	113.0 INCHES
WIDTH	36.25 INCHES
HEIGHT (with stacking posts up)	24.75 INCHES
STACKING HEIGHT (3 high)	67.75 INCHES
WEIGHT (empty)	510.0 LBS.
WEIGHT (loaded-2 weapons)	1462.0 LBS.

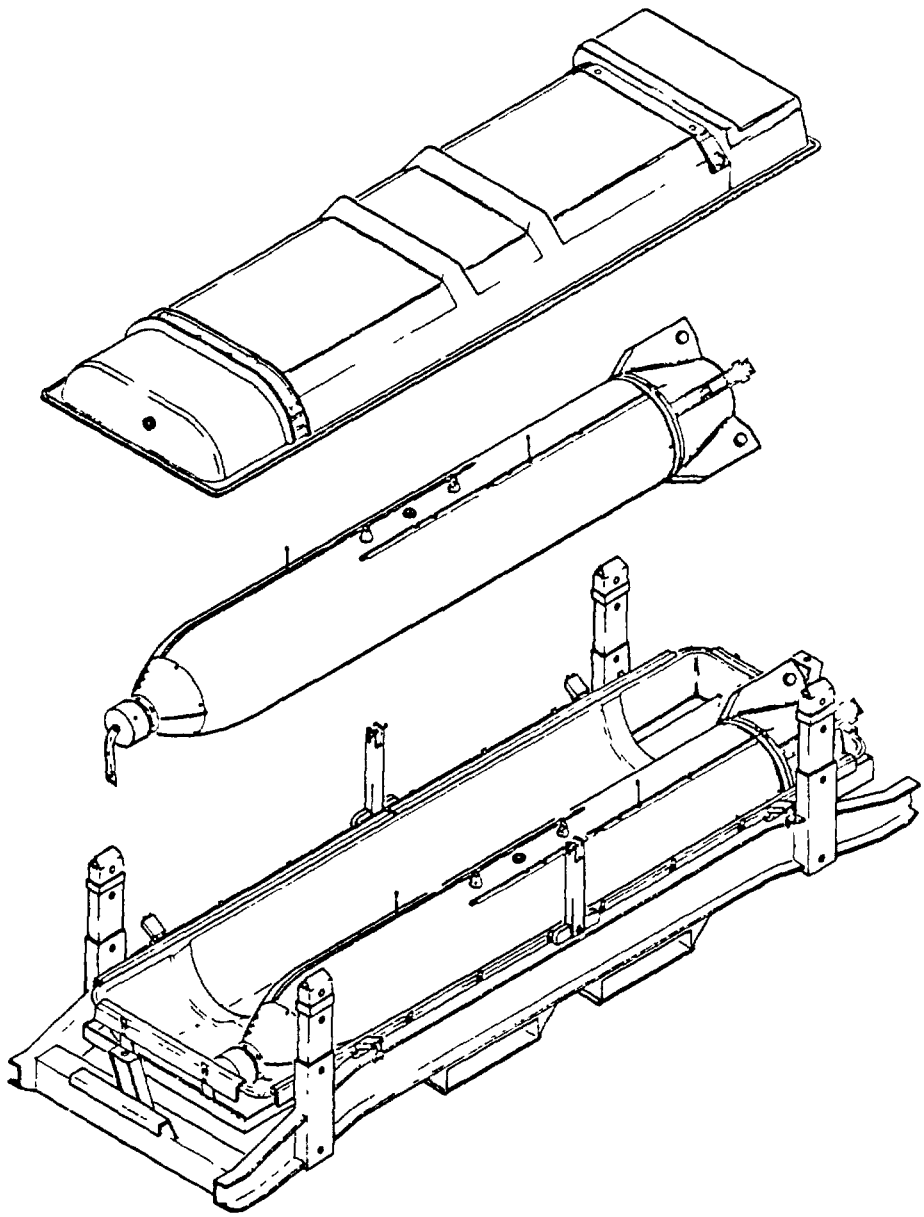


FIGURE 6. CRADLE, WEAPON, MK 18 MOD 0

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*

SPECIFICATION	
MIL-C-81585(AS)	CRADLES, WEAPON, SHIPPING AND STORAGE, ML 12, MK 13, MK 14, MK 15, MK 16 AND MK 18

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED *(Printed or typed name and activity)*	DATE

DD FORM 1 NOV 64 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

Weapons Engineering Standardization Office (Code X)
Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:45Z
Check the source to verify that this is the current version before use.